

CSCE 222

## Homework 3

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### Contents

|       |                                                                                                                                                                                                 |   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 0.1   | 12 . . . . .                                                                                                                                                                                    | 2 |
| 0.1.1 | a . . . . .                                                                                                                                                                                     | 2 |
| 0.1.2 | b . . . . .                                                                                                                                                                                     | 3 |
| 0.1.3 | c . . . . .                                                                                                                                                                                     | 3 |
| 0.1.4 | d . . . . .                                                                                                                                                                                     | 3 |
| 0.2   | Question 14 - Find the recurrence relation satisfied by the sequence $\{a_n\}$ . . . . .                                                                                                        | 4 |
| 0.2.1 | a . . . . .                                                                                                                                                                                     | 4 |
| 0.2.2 | b . . . . .                                                                                                                                                                                     | 4 |
| 0.2.3 | c . . . . .                                                                                                                                                                                     | 4 |
| 0.2.4 | d . . . . .                                                                                                                                                                                     | 4 |
| 0.2.5 | e . . . . .                                                                                                                                                                                     | 5 |
| 0.2.6 | f . . . . .                                                                                                                                                                                     | 5 |
| 0.2.7 | g . . . . .                                                                                                                                                                                     | 5 |
| 0.2.8 | h . . . . .                                                                                                                                                                                     | 5 |
| 0.3   | Question 22a - An employee joined a company in 2023 with a starting salary of \$50,000. Every year this employee receives a raise of \$1000 plus 5% of the salary of the previous year. . . . . | 5 |
| 0.3.1 | a . . . . .                                                                                                                                                                                     | 5 |

|          |                                                                  |          |
|----------|------------------------------------------------------------------|----------|
| 0.4      | Question 22b - What will the salary of this employee be in 2031? | 6        |
| 0.5      | Question 32 - Find the value of each of these sums. . . . .      | 6        |
| 0.5.1    | a . . . . .                                                      | 6        |
| 0.5.2    | b . . . . .                                                      | 6        |
| 0.5.3    | c . . . . .                                                      | 6        |
| 0.5.4    | d . . . . .                                                      | 7        |
| <b>1</b> | <b>Section 5.1</b>                                               | <b>7</b> |
| 1.1      | Question 8 . . . . .                                             | 7        |
| 1.2      | Question 12 . . . . .                                            | 7        |
| 1.3      | Question 14 . . . . .                                            | 7        |
| 1.4      | Question 18 . . . . .                                            | 7        |

**0.1 12 - Show that the sequence  $\{a_n\}$  is a solution of the recurrence relation  $a_n = -3a_{n-1} + 4a_{n-2}$  if**

$$P(n) = a_n = -3a_{n-1} - 1 + 4a_{n-2} \text{ for all } n \geq 2$$

**0.1.1 a -  $a_n = 0$**

Basis step:  $P(2) = a_2 = 0 = -3(0) + 4(0) = 0$  which holds

Inductive step:  $p(k) \rightarrow p(k+1)$

Assume  $p(k)$  holds for  $k \geq 2$ ,  $a_k = 0, a_{k-1} = 0$

$$\text{LHS: } a_{k+1} = 0$$

$$\text{RHS: } -3a_k + 4a_{k-1}$$

$$-3(0) + 4(0) = 0$$

LHS = RHS therefore  $p(k) \rightarrow p(k+1)$  holds for all  $n \geq 2$  by induction.

**0.1.2   b -  $a_n = 1$** 

Basis step:  $P(2) = a_2 = 1 = -3(1) + 4(1) = 1$  which holds

Inductive step:  $p(k) \rightarrow p(k+1)$

Assume  $p(k)$  holds for  $k \geq 2$ ,  $a_k = 1, a_{k-1} = 1$

LHS:  $a_{k+1} = 1$

RHS:  $-3a_k + 4a_{k-1}$

$-3(1) + 4(1) = 1$

LHS = RHS therefore  $p(k) \rightarrow p(k+1)$  holds for all  $n \geq 2$  by induction.

**0.1.3   c -  $a_n = (-4)^n$** 

Basis step:  $P(2) = a_2 = (-4)^2 = 16 = -3((-4)^1) + 4((-4)^0) = 12 + 4 = 16$   
which holds

Inductive step:  $p(k) \rightarrow p(k+1)$

Assume  $p(k)$  holds for  $k \geq 2$ ,  $a_k = -4^k, a_{k-1} = -4^{k-1}$

LHS:  $a_{k+1} = (-4)^{k+1}$

RHS:  $-3a_k + 4a_{k-1}$

$-3((-4)^k) + 4((-4)^{k-1})$

$-3((-4)^k) + 4((-4)^k(-4)^{-1})$

$-3((-4)^k) + ((-4)^k(-1))$

$-4(-4)^k$

$(-4)^1((-4)^k)$

$((-4)^{k+1})$

LHS = RHS therefore  $p(k) \rightarrow p(k+1)$  holds for all  $n \geq 2$  by induction.

**0.1.4   d -  $a_n = 2(-4)^n + 3$** 

Basis step:  $P(2) = a_2 = 2(-4)^2 + 3 = 35 = -3(2(-4)^1 + 3) + 4(2(-4)^0 + 3) =$   
 $15 + 20 = 35$  which holds

Inductive step:  $p(k) \rightarrow p(k+1)$

Assume  $p(k)$  holds for  $k \geq 2$ ,  $a_k = 1, a_{k-1} = 1$

LHS:  $2(-4)^{k+1} + 3$

$$\text{RHS: } -3a_k + 4a_{k-1}$$

$$-3(2(-4)^k + 3) + 4(2(-4)^{k-1} + 3)$$

$$(-6(-4)^k + -9) + (8(-4)^k(-4)^{-1} + 12)$$

$$(-6(-4)^k + -9) + ((-4)^k(-2) + 12)$$

$$(-6(-4)^k) + (-2(-4)^k) + 3$$

$$(-8(-4)^k) + 3$$

$$((2(-4)^1)(-4)^k) + 3$$

$$2(-4)^{k+1} + 3$$

LHS = RHS therefore  $p(k) \rightarrow p(k+1)$  holds for all  $n \geq 2$  by induction.

## 0.2 Question 14 - Find the recurrence relation satisfied by the sequence $\{a_n\}$

### 0.2.1 a - $a_n = 0$

$a_n = 0(a_{n-1})$  for all  $n \geq 1$ , where  $a_0 = 0$

### 0.2.2 b - $a_n = 1$

$a_n = (a_{n-1})$  for all  $n \geq 1$ , where  $a_0 = 1$

### 0.2.3 c - $a_n = 2^n$

$a_n = 2(a_{n-1})$  for all  $n \geq 1$ , where  $a_0 = 1$

### 0.2.4 d - $a_n = 4^n$

$a_n = 4(a_{n-1})$  for all  $n \geq 1$ , where  $a_0 = 1$

**0.2.5 e -**  $a_n = n4^n$

$a_n = 8(a_{n-1}) - 16(a_{n-2})$  for all  $n \geq 2$ , where  $a_0 = 0, a_1 = 4$

**0.2.6 f -**  $a_n = 2(4^n) + 3n4^n$

$a_n = 8(a_{n-1}) - 16(a_{n-2})$  for all  $n \geq 2$ , where  $a_0 = 0, a_1 = 4$

**0.2.7 g -**  $a_n = (-4)^n$

$a_n = -4(a_{n-1})$  for all  $n \geq 1$ , where  $a_0 = 1$

**0.2.8 h -**  $a_n = n^2 4^n$

$a_n = 12(a_{n-1}) - 48(a_{n-2}) + 64(a_{n-3})$  for all  $n \geq 3$ , where  $a_0 = 0, a_1 = 4, a_2 = 64$

**0.3 Question 22a - An employee joined a company in 2023 with a starting salary of \$50,000. Every year this employee receives a raise of \$1000 plus 5% of the salary of the previous year.**

**0.3.1 a - Set up a recurrence relation for the salary of this employee  $n$  years after 2023.**

$a_0 = 50,000$

$a_n = a_{n-1} + 1,000 + 0.05a_{n-1}$

$a_n = 1.05a_{n-1} + 1,000$  for all  $n \geq 1$ , where  $a_0 = 50,000$

## 0.4 Question 22b - What will the salary of this employee be in 2031?

$$n = 2031 - 2023 = 8$$

$$a_n = 1.05a_{n-1} + 1,000 \text{ for all } n \geq 1, \text{ where } a_0 = 50,000$$

$$a_1 = 1.05a_0 + 1,000 = 53,500 \text{ in 2024}$$

$$a_2 = 1.05a_1 + 1,000 = 57,175 \text{ in 2025}$$

$$a_3 = 1.05a_2 + 1,000 = 61,033.75 \text{ in 2026}$$

$$a_4 = 1.05a_3 + 1,000 = 65,085.44 \text{ in 2027}$$

$$a_5 = 1.05a_4 + 1,000 = 69,339.71 \text{ in 2028}$$

$$a_6 = 1.05a_5 + 1,000 = 73,806.70 \text{ in 2029}$$

$$a_7 = 1.05a_6 + 1,000 = 78,497.04 \text{ in 2030}$$

$$a_8 = 1.05a_7 + 1,000 = 83,421.89 \text{ in 2031}$$

Thus, the salary in 2031 would be \$83,421.89

## 0.5 Question 32 - Find the value of each of these sums.

### 0.5.1 a- $\sum_{j=0}^8 1 + (-1)^j$

$$\sum_{j=0}^8 1 + (-1)^j = \sum_{j=0}^8 1 + \sum_{j=0}^8 (-1)^j$$

$$\sum_{j=0}^8 1 + (-1)^j = 9(1) + 1 + 4(-1 + 1) = 10$$

### 0.5.2 b- $\sum_{j=0}^8 (3^j - 2^j)$

$$\sum_{j=0}^8 (3^j - 2^j) = \sum_{j=0}^8 (3^j) - \sum_{j=0}^8 (2^j)$$

Using the geometric series formula:

$$\sum_{j=0}^8 (3^j - 2^j) = \left( \frac{3^0(1-3^9)}{1-3} \right) - \left( \frac{2^0(1-(2)^9)}{1-2} \right)$$

$$\sum_{j=0}^8 (3^j - 2^j) = (9841 - 511) = 9330$$

### 0.5.3 c- $\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j)$

Using the geometric series formula:

$$\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j) = 2 \left( \frac{3^0(1-3^9)}{1-3} \right) + 3 \left( \frac{2^0(1-(2)^9)}{1-2} \right)$$

using part b,

$$\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j) = 2(9841) + 3(511) = 21215$$

**0.5.4 d-**  $\sum_{j=0}^8 (2^{j+1} - 2^j)$

$$\sum_{j=0}^8 (2^{j+1} - 2^j) = \sum_{j=0}^8 (2^j 2 - 2^j)$$

$$\sum_{j=0}^8 (2^{j+1} - 2^j) = (2 \sum_{j=0}^8 2^j) - (\sum_{j=0}^8 (2^j)) = (\sum_{j=0}^8 (2^j))$$

Using the geometric series formula:

$$\sum_{j=0}^8 (2^{j+1} - 2^j) = \frac{1-2^9}{1-2} = 511$$

## 1 Section 5.1

### 1.1 Question 8

### 1.2 Question 12

### 1.3 Question 14

### 1.4 Question 18